Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

Method 1 / 方法 1 [NT07-M01]

A fraction problem is usually controlled by reduction, not by decimal approximation. Keep denominators exact, identify the cancellation gcd, and separate distinct values from different representations of the same value.

分数题通常由约分控制，而非小数近似控制。保留精确分母，确定约分公因数，并区分不同数值与同一数值的不同表示。

Convert a repeating block to an exact fraction

将循环节转为精确分数

Procedure

1. If a length-k block represents integer A in base b, then $0.\overline{A}_b = A/(b^k-1)$.

2. Leading zeroes in the block are legal; a shorter actual period may still be represented by the full block.

3. For sums over digit permutations, count how often each digit occupies each position before reducing fractions.

通用步骤

1. 若底数 b 下长度 k 的块表示整数 A，则 $0.\overline{A}_b = A/(b^k-1)$。

2. 循环块开头允许零，实际周期较短者仍可用完整循环块表示。

3. 对数位排列求和时，先统计每个数字在每个位置的出现次数，再约分。

> Example. Find the sum of $0.\overline{12}$, $0.\overline{21}$ and $0.\overline{03}$.

> 例题：求 $0.\overline{12}, 0.\overline{21}, 0.\overline{03}$ 之和。

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

Method 1 / 方法 1 [NT07-M01] (continued / 续)

Worked solution

1. The common denominator for a two-digit block is 99.

2. The numerators are $12, 21, 3$, including the leading-zero block 03.

3. The sum is $36/99 = 4/11$.

例题解析

1. 两位循环块的共同分母为 99。

2. 分子为 $12, 21, 3$，其中包括开头为零的循环块 03。

3. 和为 $36/99 = 4/11$。

The block length is fixed even when its integer value has fewer digits.

即使循环块的整数值位数较少，循环块长度仍固定。

Practice / 配套练习: AIME-2006-I-Q06 (WB p. 596), AIME-2014-II-Q04 (WB p. 594)

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

Method 2 / 方法 2 [NT07-M02]

Control reduction with a gcd and denominator prime structure

用最大公因数与分母质因数结构控制约分

Procedure

1. Write $r = a/b$ in lowest terms and compute explicitly the gcd introduced by a multiplier or common denominator.

2. After multiplying by c, the cancellation is $\gcd(c, b)$ because $\gcd(a, b) = 1$.

3. For an integrality condition, every prime-power in the reduced denominator must be supplied by the numerator; use valuation bounds to prove the first possible index.

4. With a fixed numerator-plus-denominator sum, replace their gcd by the gcd with that fixed sum.

通用步骤

1. 将 $r = a/b$ 写成既约形式，并明确计算乘数或公分母引入的最大公因数。

2. 乘以 c 后，约分公因数为 $\gcd(c, b)$，因为 $\gcd(a, b) = 1$。

3. 整数条件要求分子提供既约分母的每个质数幂，用赋值界证明最早可能下标。

4. 若分子分母之和固定，可将其公因数改写为与固定和的公因数。

> Example. A reduced fraction a/b becomes a fraction with denominator $b/6$ after multiplication by 18. Characterize b.

> 例题：既约分数 a/b 乘以 18 后，最简分母为 $b/6$。刻画 b。

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

Method 2 / 方法 2 [NT07-M02] (continued / 续)

Worked solution

1. The new denominator is $b/\gcd(18,b)$, so the condition is $\gcd(18,b)=6$.

2. Since $18=2\cdot 3^2$, this requires $v_2(b)\ge 1$ and $v_3(b)=1$; other primes are unrestricted.

3. Equivalently $b=6t$ with $3\nmid t$. Any a coprime to such a b realizes the stated cancellation.

例题解析

1. 新分母为 $b/\gcd(18,b)$，所以条件为 $\gcd(18,b)=6$。

2. 因 $18=2\cdot 3^2$，须 $v_2(b)\ge 1, v_3(b)=1$，其他质数不受限制。

3. 等价于 $b=6t$ 且 $3\nmid t$；任取与该 b 互质的 a 均实现所述约分。

Compute the cancellation gcd before comparing reduced numerators and denominators.

比较最简分子分母前，先算约分公因数。

Practice / 配套练习: AIME-2006-II-Q14 (WB p. 481), AIME-2014-I-Q03 (WB p. 603), AIME-2023-II-Q05 (WB p. 595)

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

Method 3 / 方法 3 [NT07-M03]

Characterize the possible reduced numerators using divisor denominators

用约数分母刻画可能的最简分子

Procedure

1. For fractions k/N with $1 \le k \le N$, every reduced denominator is a divisor D of N.

2. A positive numerator m occurs exactly when some such D satisfies $m \le D$ and $\gcd(m, D) = 1$.

3. Partition m by which primes of N divide it. In each class choose the largest divisor of N avoiding those primes, then count under that bound.

通用步骤

1. 对 $1 \le k \le N$ 的分数 k/N，每个最简分母都是 N 的约数 D。

2. 正分子 m 出现，当且仅当某个这样的 D 满足 $m \le D, \gcd(m, D) = 1$。

3. 按 N 的哪些质数整除 m 分类。每类选避开这些质数的最大约数作分母，再在该上界下计数。

> Example. Among $k/12$ for $1 \le k \le 12$, which positive reduced numerators occur?

> 例题：在 $1 \le k \le 12$ 的 $k/12$ 中，会出现哪些正的最简分子?

Worked solution

1. If m is coprime to 6, choose denominator 12; the allowed numerators are $1, 5, 7, 11$.

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

Method 3 / 方法 3 [NT07-M03] (continued / 续)

2. If $2 \mid m$ but $3 \nmid m$, the largest coprime divisor denominator is 3, giving only $m = 2$.

3. If $3 \mid m$ but $2 \nmid m$, denominator 4 permits only $m = 3$. A numerator divisible by both primes is too large for denominator one. Thus the set is $\{1, 2, 3, 5, 7, 11\}$.

例题解析

1. 若 m 与 6 互质，可取分母 12，得到 $1, 5, 7, 11$。

2. 若 $2 \mid m$ 而 $3 \nmid m$，最大互质约数分母为 3，只得到 $m = 2$。

3. 若 $3 \mid m$ 而 $2 \nmid m$，分母 4 只允许 $m = 3$。同时被两质数整除的分子对分母一而言太大。故集合为 $\{1, 2, 3, 5, 7, 11\}$。

Count distinct reduced numerators, not unreduced blocks or fraction representations.

应数不同最简分子，而非未约分循环块或分数表示。

Practice / 配套练习: AIME-2022-I-Q13 (WB p. 598)

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

Method 4 / 方法 4 [NT07-M04]

Use a decimal interval to bound a reduced denominator

用小数区间限制最简分母

Procedure

1. Turn a fixed length-k prefix A into $[A/10^k, (A+1)/10^k)$.

2. If the prefix begins later, shift the decimal and take the fractional part; its reduced denominator divides the original denominator.

3. Compare with a nearby simple fraction and exploit a nonzero integer cross-product difference to get a lower bound.

通用步骤

1. 将固定长度 k 的前缀 A 转为 $[A/10^k, (A+1)/10^k)$。

2. 若前缀在较后位置开始，移动小数点并取小数部分；其最简分母整除原分母。

3. 与附近简单分数比较，并利用非零整数交叉乘积差得到下界。

> Example. Prove no reduced fraction with denominator at most 10 lies in $[0.501, 0.510)$.

> 例题：证明分母不超过 10 的既约分数不可能位于 $[0.501, 0.510)$。

Worked solution

1. Such a fraction m/n is strictly above $1/2$, so $2m - n \geq 1$.

2. Its distance above $1/2$ is at least $1/(2n) \geq 1/20 = 0.05$.

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

Method 4 / 方法 4 [NT07-M04] (continued / 续)

3. But the stated interval requires that distance be below 0.010. This contradiction excludes every such denominator.

例题解析

1. 这样的分数 m/n 严格大于 $1/2$，所以 $2m - n \geq 1$。

2. 它与 $1/2$ 的距离至少为 $1/(2n) \geq 1/20 = 0.05$。

3. 而给定区间要求该距离小于 0.010，矛盾，故排除所有这些分母。

> The strict upper endpoint is essential when the next decimal prefix begins there.

> 右端点正是下一小数前缀开始处，严格性不可忽略。

Practice / 配套练习: AIME-2003-I-Q14 (WB p. 597)

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

Method 5 / 方法 5 [NT07-M05]

Locate the unique least valuation in a rational sum

定位有理和中唯一最低赋值项

Procedure

1. Reduce each summand and compare its exponent of a chosen prime in the denominator.

2. Clear the largest denominator power. If exactly one resulting numerator is a unit modulo the prime, the total cannot cancel that prime.

3. Compute the exact surviving denominator exponent; use a factorial valuation formula when needed.

通用步骤

1. 约分每项，并比较所选质数在分母中的指数。

2. 清除最大分母幂。若所得分子中恰有一项模该质数为单位，则总和不能约去该质数。

3. 算出保留的精确分母指数，必要时使用阶乘赋值公式。

> Example. Find the power of 2 in the reduced denominator of $1/2+3/8+5/32$.

> 例题：求 $1/2+3/8+5/32$ 的最简分母中 2 的指数。

Worked solution

1. The common denominator is 32, giving numerator $16+12+5$.

2. Only the last contribution is odd, so the total numerator 33 is odd.

3. No two-factor cancellation occurs: the reduced denominator remains $32=2^5$.

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

Method 5 / 方法 5 [NT07-M05] (continued / 续)

例题解析

1. 共同分母为 32，分子为 $16+12+5$。

2. 只有最后一项贡献为奇数，故总分子 33 为奇数。

3. 不能约去任何因子二，所以最简分母仍为 $32=2^5$。

If two terms share the lowest valuation, cancellation may occur and needs another analysis.

若两项共享最低赋值，则可能相消，须另作分析。

Practice / 配套练习: AIME-2009-II-Q07 (WB p. 475)